

DMU MODEL 2 FOR CS

1. DMA interface unit eliminates the need to use CPU registers to transfer data from
 - ☐ A) I/O units to memory
 - ☐ B) Memory address register to Memory buffer register
 - ☐ C) Memory buffer register to Memory address register
 - ☐ D) Memory to I/O units
2. The entity integrity rule states that _____.
 - ☐ A) no primary key attribute can be null.
 - ☐ B) referential integrity must be maintained across all entities.
 - ☐ C) each entity must have a primary key.
 - ☐ D) a primary key must have only one attribute
3. One is not an advantage of using Multiprocessor Operating System.
 - ☐ A) Increase Throughput
 - ☐ B) Increase Turnaround time
 - ☐ C) Decrease Waiting time of a process
 - ☐ D) Decrease response time of the process
4. In which types of distribute database system that each site may run different database system but the data access is managed through a single conceptual schema.
 - ☐ A) Federated database
 - ☐ B) Multidatabase
 - ☐ C) Homogeneous database
 - ☐ D) Standalone Database
5. Which of the following is not a type of graph in computer science?
 - ☐ A) undirected graph
 - ☐ B) bar graph
 - ☐ C) directed graph
 - ☐ D) weighted graph
6. A command that is used to Run active directory is
 - ☐ A) dcpro
 - ☐ B) gpupdate

- ☐ C) dcpromo
- ☐ D) dcpro /add ad

7. Which one is wrong about deferred update recovery scheme?

- ☐ A) No undo is required to recover
- ☐ B) It uses Redo to reboot from failure
- ☐ C) Changes before reach commit point should discarded
- ☐ D) The update are first recorded on the log and on the disk by force writing before the database is updated

8. What will be the output of the following program? #include using namespace std; void sayHello(); { cout << "Hello World" } int main() { sayHello(); return 0; }

- ☐ A) compile timer error
- ☐ B) Hello World
- ☐ C) sayHello
- ☐ D) Hello

9. ____ is a header file used in C++ that handles input and output functions.

- ☐ A) conio.h
- ☐ B) stdio.h
- ☐ C) string.h
- ☐ D) iostream.h

10. Which one of the following facilitates transfer of bulk data from hard disk to main memory with highest throughput?

- ☐ A) Interrupt driven I/O transfer
- ☐ B) Polling based I/O transfer
- ☐ C) DMA based I/O transfer
- ☐ D) Programmed I/O transfer

11. Which one of the following is at most one variable on the right side of the production, must be the left-most symbol.

- ☐ A) Right-linear grammar
- ☐ B) Left-linear grammar.
- ☐ C) regular languages
- ☐ D) Context free language

12. Suppose you are a software engineer and focusing on producing a model of the system, which is correct, complete, consistent, and verifiable is called_____.

- ☐ A) Design model

- ☐ B) Analysis Model
- ☐ C) Testing model
- ☐ D) Requirement gathering

13. Which requirement elicitation techniques are used to gather the requirements using a scenario-based technique in the UML which identifies the actors in an interaction and which describe the interaction itself?

- ☐ A) Scenarios
- ☐ B) Documents
- ☐ C) Ethnography
- ☐ D) Use case

14. A mechanism used for a subclass to implement a super class method based on its requirement is?

- ☐ A) Overriding
- ☐ B) Overloading
- ☐ C) Polymorphism
- ☐ D) Encapsulation

15. It holds all of the information needed to create users, groups, computers within active directory

- ☐ A) Schema
- ☐ B) Forest
- ☐ C) Tree
- ☐ D) Domain

16. One of the following searching strategy is not grouped under informed/heuristic search to solve problems.

- ☐ A) Greedy search
- ☐ B) Iterative Deeping search
- ☐ C) Constraint satisfaction search
- ☐ D) Iterative improvement search

17. Which one of the command that used to activate the changes in active directory?

- ☐ A) gpupdate
- ☐ B) dcpromo
- ☐ C) gpchange
- ☐ D) change

18. Which one of the following is true about a standard Turing machine is capable of

accepting some of the languages and grammars respectively?

- ☒ A) recursively enumerable language, unrestricted grammars
- ☐ B) Context sensitive language, Context free grammars
- ☐ C) Context free language, regular grammars
- ☐ D) Pushdown automata, regular expression

19. Which of the following is the task of Unsupervised learning technique of artificial intelligence?

- ☐ A) Classification
- ☐ B) Regression
- ☒ C) Clustering
- ☐ D) Take action based of reward & punishment

20. Processes Arrival Time Burst Time P1 0 10 P2 2 6 P3 4 1 P4 5 3 The average waiting time and turnaround time of the processes using pre-emptive Shortest Job First algorithm is _____ and _____ respectively.

- ☐ A) 2.5 and 6.5
- ☒ B) 3 and 7
- ☐ C) 4.5 and 8
- ☐ D) 3.5 and 8.5

21. Which UML diagram describes the interactions among classes in terms of an exchange of messages over time?

- ☐ A) Use case diagram
- ☒ B) Sequence diagram
- ☐ C) Class diagram
- ☐ D) Deployment diagram

22. Which one is true about scheduling algorithms?

- ☐ A) First-Come-First-Served Scheduling (FCFSS) algorithm is non-preemptive algorithm and is efficient algorithm.
- ☒ B) SJF algorithm can be either preemptive or non-preemptive algorithm and it is desirable algorithm for a time-sharing system.
- ☐ C) Round-robin (RR) scheduling algorithm is designed especially for time sharing systems but it is not fairest algorithm for short jobs
- ☐ D) Priority scheduling algorithm doesn't consider the importance of processes.

23. Which one of the following is odd?

- ☐ A) Update
- ☐ B) Delete
- ☒ C) Drop

☐ D) Insert

24. Which of the following language translator program is used in C++?

☐ A) Assembler

☐ B) Both Interpreter and Compiler

☒ C) Compiler

☐ D) Interpreter

25. Which one of the following is more related with data hiding?

☐ A) Polymorphism

☐ B) Inheritance

☒ C) Encapsulation

☐ D) overloading

26. which of the following correct recurrence with time complexity of worst case of bubble sort Algorithm?

☒ A) Recurrence is $T(n) = T(n-2) + O(n)$ and time complexity is $O(n^2)$

☐ B) Recurrence is $T(n) = T(n-1) + O(n)$ and time complexity is $O(n)$

☐ C) Recurrence is $T(n) = 2T(n/2) + O(n)$ and time complexity is $O(n \log n)$

☐ D) Recurrence is $T(n) = T(n/10) + T(9n/10) + O(n)$ and time complexity is $O(n \log n)$

27. Which of the following is true in the case of the software design process?

☐ A) Analysis occurs at the start of product design with a product idea

☒ B) Analysis occurs at the end of engineering design with the SRS

☐ C) Product design resolution produces the design document

☐ D) Engineering design resolution produces the SRS

28. Which of the following statement is correct about the global variable?

☐ A) Global variables can only declare inside the ".h" file.

☐ B) A variable does not have global scope

☒ C) A variable defined outside the function or block is known as a global variable.

☐ D) variable defined inside the function or block is known as a global variable.

29. The ability to use the same name for two or more methods in a class is?

☒ A) Overriding

☐ B) Overloading

☐ C) Polymorphism

☐ D) Encapsulation

30. Which types of tasks is performed by intelligent system without intelligence
- ☐ A) Use natural language fluently
 - ☒ B) Calculate something
 - ☐ C) Store and retrieve information from memory
 - ☐ D) None
31. Which ARIES recovery phase reads the log from the last recorded checkpoint to identify any dirty pages and active transactions at the time of the crash?
- ☐ A) Redo phase
 - ☐ B) Undo phase
 - ☒ C) Analysis phase
 - ☐ D) Dirty page
32. In process instruction cycle the step during which the operations specified by the instruction are executed is known as _____.
- ☐ A) Fetch Cycle
 - ☐ B) Indirect Cycle
 - ☒ C) Execute Cycle
 - ☐ D) Interrupt Cycle
33. Which one of the following data base design modelling is important to represent the conceptual view of the database?
- ☒ A) ER diagram model
 - ☐ B) Hierarchical model
 - ☐ C) Network model
 - ☐ D) Relational data model
34. A process is waiting for the resource held by the second process, which is waiting for the resource held by the third process and so on, till the last process is waiting for a resource held by the first process. This chain of deadlock characterization is:
- ☒ A) Circular wait
 - ☐ B) Hold and wait
 - ☐ C) Mutual exclusion
 - ☐ D) No pre-emption
35. A hierarchical structure made up of multiple domains that trust each other is called?
- ☒ A) Forest
 - ☐ B) Schema
 - ☐ C) Site
 - ☐ D) Organizational Unit

36. Which one of the following from short hands is represents zero or one of zero or one of the regular expression?

- ☐ A) $r^+ = rr^*$
- ☐ B) $r^* = r^+ \mid \epsilon$
- ☒ C) $r^? = r \mid \epsilon$
- ☐ D) $r^* = r^*r^*$

37. Which of the following is correct about memory-mapped I/O

- ☐ A) The I/O devices have a separate address space
- ☐ B) The memory and I/O devices have an associated address space
- ☒ C) The I/O devices and the memory share the same address space
- ☐ D) Part of the memory is specifically set aside for the I/O operation

38. Which one of the following is the class of decision problems that have polynomial-time deterministic algorithms?

- ☐ A) class NP
- ☒ B) class P
- ☐ C) Non polynomial-time
- ☐ D) Nondeterministic algorithms

39. Which one of the following is the functions of modern operating system connection with process management?

- ☒ A) Managing process and threads
- ☐ B) Managing memory partition
- ☐ C) Allocation & de-allocation of devices to processes
- ☐ D) Managing files and directories

40. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____ integrity constraint.

- ☒ A) Referential
- ☐ B) Primary
- ☐ C) Referencing
- ☐ D) Specific

41. If you want to search the key element in 'n' number of array elements by using a binary search algorithm and the searched key element is greater than mid element, then what is the position of top and bottom element?

- ☐ A) Top is n-1 and bottom is mid-1

- ☐ B) Top is $n-1$ and bottom is $mid + 1$
- ☐ C) Top is $n+1$ and bottom is $mid + 1$
- ☐ D) Top is $n+1$ and bottom is $mid - 1$

42. What is time complexity of fun ()? `int fun(int n) { int count = 0; for (int i = n; i > 0; i /= 2) for (int j = 0; j < i; j++) count += 1; return count; }`

- ☐ A) $O(n^2)$
- ☐ B) $O(n \cdot \log(n))$
- ☒ C) $O(n)$
- ☐ D) $O(n \cdot \log(n \cdot \log(n)))$

43. A user logs on to her account and is able to make changes to her desktop, like adding a background and screen saver. However, when user logs off and logs back on, the desktop changes to its original configuration. What type of user profile does this user have?

- ☐ A) Roaming profile
- ☒ B) Local profile
- ☐ C) Mandatory profile
- ☐ D) Managing profile

44. Which of the following is a dynamic programming problem?

- ☒ A) Longest Common Subsequence
- ☐ B) Depth First Search
- ☐ C) Binary Search
- ☐ D) Breadth First Search

45. Which one of the following is not true about a regular expression?

- ☐ A) $\emptyset$ denotes language $\emptyset$
- ☐ B) ε denotes language $\{\varepsilon\}$
- ☐ C) a denotes language $\{a\}$, for all $a \in \Sigma$.
- ☒ D) ab denotes language $\{acb\}$, for all $a, c \in \Sigma$.

46. You tried to reset a password, but received a message that your password does not meet the password complexity requirements. What might be the problem?

- ☐ A) The maximum password age is set to 1
- ☒ B) The user password is not complex enough
- ☐ C) The minimum password length is less than 6 characters
- ☐ D) Password never expires

47. Let us suppose that in a distributed database, during a transaction T_1 , one of the sites, say S_1 , is failed. When recovers, the site S_1 has to check its log file (log based recovery) to decide

the next move on the transaction T1. If the log contains a record, what the site S1 has to do?

- ☐ A) Need not do anything
- ☐ B) Perform Undo
- ☒ C) Perform Redo
- ☐ D) Abort the transaction

48. Which of the following algorithms is used to find the minimum number of coins needed to make change for a given amount?

- ☒ A) Greedy algorithm
- ☐ B) Depth First Search
- ☐ C) Dijkstra's shortest path algorithm
- ☐ D) Breadth First Search

49. Which one of the following is defining software and hardware architecture, components, modules, interfaces, and data to enable a system to satisfy a set of well-specified operational requirements?

- ☐ A) System analysis
- ☐ B) Implementation
- ☒ C) System design
- ☐ D) Requirement Elicitation

50. Consider the following C++ programming function `void fun (int x) { node *temp=new node; temp->data=x; temp->next=NULL; if(head==NULL) head = temp; else { node *temp2 = head; while(temp2->next!=NULL) { temp2 = temp2->next; } temp2->next= temp; } }` After we call the above c++ programs how many linked list nodes is/are modified?

- ☐ A) One node modifies
- ☒ B) Two node modifies
- ☐ C) Three node modifies
- ☐ D) Four node modifies

51. Which unit of CPU perform arithmetic operations such as addition, subtraction, multiplication, division over data received from memory and compare numbers (greater than, equal to, or less than)?

- ☒ A) ALU
- ☐ B) Control Unit
- ☐ C) ALU and Control Unit
- ☐ D) Input/Output Unit

52. All of the following are the main goals of normalization EXCEPT

- ☐ A) Minimize data redundancy.
- ☐ B) Simplify the enforcement of referential integrity.

- ☐ C) Maximize storage space.
- ☐ D) make it easier to maintain data

53. One of the following is not an objective of operating system.

- ☐ A) An OS makes a computer more convenient to use.
- ☐ B) An OS allows the computer system resources to be used in an efficient manner.
- ☐ C) An OS should be constructed in such a way as to permit the effective development, testing, and introduction of new system functions without interfering with service.
- ☐ D) OS Keeps track of time and resources used by various jobs and users

54. Which of the following is/ are true about queue type data structure after adding additional element on queue?

- ☐ A) Front and rear pointer increment by one
- ☐ B) Rear pointer increment by one and front pointer is not
- ☐ C) Front pointer increment by one and rear is not
- ☐ D) All

55. If transaction had not committed at failure time, recovery manager has to perform ____ to recover.

- ☐ A) Redo operation
- ☐ B) Undo operation
- ☐ C) The change should be saved
- ☐ D) Non-recoverable

56. Which of the following is/are true about stack type of data structure?

- ☐ A) It is a linear types of data structure in which insertion and deletion performed in both ends.
- ☐ B) It is a non-linear types of data structure in which insertion and deletion performed in both ends
- ☐ C) It is a primitive types of data structure in which follows last-in first-out operation principle.
- ☐ D) It is a non-primitive types of data structure in which follows last-in first-out operation principle.

57. Which one of the following doesn't represent OOP?

- ☐ A) It is modular
- ☐ B) It is easily maintainable
- ☐ C) It is data centric
- ☐ D) It is faster

58. Which one uses unlimited memory as an infinite tape which can be read/written and moved to left or right?

- ☐ A) Pushdown Automata
- ☐ B) Finite Automata
- ☒ C) Turing Machines
- ☐ D) Linear bound automata.

59. of the following can be defined as data structures that allow storing a group of data of the same datatype.

- ☒ A) Array
- ☐ B) Structure
- ☐ C) String
- ☐ D) Union

60. Execution of C++ program starts from?

- ☐ A) user-defined function
- ☐ B) header file
- ☒ C) main function
- ☐ D) void function

61. Unlike RDMS, the object oriented model uses _____ to represent real world entities.

- ☒ A) Object
- ☐ B) Class
- ☐ C) Table
- ☐ D) Constructor

62. Which one is wrong about object-oriented database?

- ☒ A) It uses instance variable instead of attribute
- ☐ B) Sometimes object id can be modified
- ☐ C) It can store both the value and the operation (method) of the object
- ☐ D) It is not possible to use SQL commands

63. The process of translating a task into a series of commands that the computer will use to perform that task is _____.

- ☒ A) Programming
- ☐ B) Project design
- ☐ C) Installation
- ☐ D) System analysis

64. 2NF and 3NF are identical if _____.

- ☐ A) There are only one primary key in a relation and there is no a none keys attribute determines the other
- ☐ B) There are more than one primary key and the none primary key attribute depends on either of the primary key.
- ☐ C) There are more than one primary key and the none primary key attribute depends on both of the primary key.
- ☐ D) There are more than one primary key and there is one determinant

65. Which one of the following enables a parent class reference to refer to a child class object?

- ☐ A) Overriding
- ☐ B) Overloading
- ☐ C) Encapsulation
- ☐ D) Polymorphism

66. Which one of the following is used to create new classes from existing ones?

- ☐ A) Inheritance
- ☐ B) Encapsulation
- ☐ C) Polymorphism
- ☐ D) Overriding

67. Which of the following algorithms is NOT a divide & conquer algorithm by nature?

- ☐ A) Euclidean algorithm to compute the greatest common divisor
- ☐ B) Heap Sort
- ☐ C) Cooley-Tukey fast Fourier transform
- ☐ D) Quick Sort

68. Which components of intelligence is defined as a set of processes that enables the agent to provide basis for judgement, making decisions, and prediction.

- ☐ A) Perception
- ☐ B) Problem solving
- ☐ C) Reasoning
- ☐ D) Learning

69. Which of one of the following is not correct about knowledge engineering

- ☐ A) It is the process extracting knowledge from human knowledge engineer
- ☐ B) It is the process of choosing and problem solving strategies
- ☐ C) It is the process of choose knowledge representation formalism
- ☐ D) Its ultimate goal is design and develop knowledge base

70. If the minimum password Age on a Windows system is set to zero, what does that mean?

- ☒ A) The user can use a blank password
- ☐ B) Password complexity requirement is enabled
- ☐ C) The password never expires
- ☐ D) The account is unlock manually by the administrator